Lookit — Peak Season Ops: Bottlenecks, Workers & Home PCs

Product: Lookit (face-search for event photos)
Stack: VPS (KVM2) + AWS Rekognition + S3 now / R2 in production
Revenue model: Prepaid photo quota only (no photo sales)
Date: 2026-10-19

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1. Can home computers run workers?

Yes — with conditions. Workers only need:

1. Same `lookit-server` code + `.env`
2. Reachable **Redis** (job queue) — usually on the VPS
3. Reachable **Postgres** — usually on the VPS
4. Internet to **S3/R2** and **AWS Rekognition**
5. Machine **left on** while jobs run

Machine	Role	Works?
VPS KVM2	API + Redis + Postgres (+ optional worker)	Yes — primary
Your PC	Extra `npm run start:worker` / `dev:worker`	Yes for bursts
Other PCs (8 GB RAM)	Extra workers	Yes

Home PC is good for: peak weekends, clearing backlog, cheap extra capacity.
Home PC is bad as only production worker: sleep/shutdown, power cuts, home IP changes, unstable Wi-Fi.

Recommended layout

```
VPS KVM2  →  API + Redis + Postgres + 1 light worker (always on)
Your PC / 8GB PCs  →  extra workers when backlog is big (optional)
```

All workers must point `REDIS_URL` and `DATABASE_URL` at the **VPS**, not `127.0.0.1` on the laptop.

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2. How to run a home / office PC as a worker

1. On VPS: expose Redis (and Postgres) only to trusted IPs, with password. Prefer VPN/Tailscale/WireGuard over opening Redis to the whole internet.
2. On the PC: clone `lookit-server`, copy production-like `.env`:
   - `REDIS_URL=redis://:PASSWORD@VPS_IP:6379`
   - `DATABASE_URL=postgres://...@VPS_IP:5432/lookit`
   - Same `AWS_*` keys, same `MEDIA_STORAGE_*` / R2 or S3
3. Run: `npm ci && npm run build && npm run start:worker`
   (or `npm run dev:worker` while developing)
4. Leave the machine awake (disable sleep).
5. When done: stop worker — VPS worker continues alone.

Security: Never expose Redis without a password. Prefer Tailscale so Redis stays private.

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3. Concurrency (BullMQ “threads”)

These are **parallel jobs**, not OS threads.

VPS KVM2 (~6 vCPU / 16 GB) — API + Redis + DB on same box

Worker	Safe mean	Max try
process-photo (sharp / CPU)	3–4	6
index-faces (network → AWS)	6–8	10–12
Total both	~8–12	~15

Leave headroom for API + Postgres.

8 GB RAM home PC (worker-only)

Worker	Safe mean	Max try
process-photo	2–3	4
index-faces	4–6	8
Total	~6–8	~10

Sharp + Chrome + OS on 8 GB: stay conservative. Close browsers if indexing hard.

Your main computer (if 16 GB+)

Similar to KVM2 worker-only: process **3–4**, index **6–10**.

Tune rule: CPU pegged → cut `process-photo`. Rekognition throttle errors → cut `index-faces`.

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4. Bottlenecks and cures

#	Problem	Symptom	Cure (budget-friendly)
1	`DAILY_FACE_CALL_CAP=5000`	Index stops	Raise cap; AWS Budget alerts
2	index-faces concurrency = 1	Multi-hour backlog	Raise to 5–10 (code config)
3	process-photo concurrency = 2	Thumbs lag	Raise to 3–6
4	Everything on one VPS	API slow under load	Keep API on VPS; add home/2nd VPS workers
5	AWS Rekognition TPS	Throttle errors	Lower concurrency or request quota raise
6	Redis / RAM	OOM, stalled jobs	Cap concurrency; monitor RAM
7	Postgres on same box	Slow queries	OK early; split DB later if needed
8	S3 egress	High bill	Move media to **R2** before peak
9	Session max 500 files	Awkward big dumps	Multiple upload sessions (already supported)
10	Manual UPI activation	Paid users wait	Faster admin / daily batch
11	Guest search before indexed	Empty results	UI: “Indexed X/Y”
12	Reindex storms	Double AWS cost	Reindex only failed; fix IAM first
13	Trial abuse	Free cost burn	One trial per user; watch alts
14	Unlimited guest search	Rekognition $	Rate limit per event / IP / day
15	Collections never deleted	Storage drip	Delete collection after retention

16	Disk full	Crash	Log rotate; clean tmp; monitor disk
17	No backups	Data loss	Nightly PG dump → R2
18	No alerts	Late discovery	Uptime ping + AWS Budget + queue depth

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5. Cost reminders (SaaS only)

- **IndexFaces** ≈ $0.001 / photo (~₹0.09)
- **SearchFacesByImage** ≈ $0.001 / search
- Face storage ≈ $0.01 / 1,000 faces / month
- Plan revenue ≈ ₹0.25–0.30 / photo quota → healthy margin if you don't reindex constantly
- `DAILY_FACE_CALL_CAP` is a **budget brake** (~$5/day at 5000 × $0.001)

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6. Peak weekend playbook

1. VPS always on: API + Redis + Postgres + small worker.
2. Friday: raise daily face cap; check AWS Budget.
3. Big uploads: start workers on your PC + 8 GB PCs (Tailscale → VPS Redis).
4. Watch: queue waiting count, worker logs (`index-faces done`), VPS CPU/RAM.
5. Monday: stop home workers; leave VPS worker on.
6. After retention: delete Rekognition collections for expired events.

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7. Quick answers

Is KVM2 enough for API? Yes for early–mid season.
Is KVM2 enough alone for huge peak index? Tight — add home workers or a 2nd VPS.
Do 8 GB PCs work as workers? Yes, mean concurrency ~6–8 total jobs.
Must photos be on S3 for Rekognition? No — worker downloads from S3/R2 and sends **bytes** to Rekognition.

Internal ops note for Lookit. Update concurrency numbers in worker source when you change defaults.